

CarTrack Pro

AI-Powered Car Service Monitoring, Vehicle Tracking & Business Analytics Platform

11

CORE MODULES

23 + SVCS

SERVICE TYPES

6

AI CAMERA FEEDS

4

ACCESS ROLES

Stack: React 18 · FastAPI · PostgreSQL · YOLOv8 · EasyOCR

Deployment: On-Premise · Docker · Nginx

Access: Web Browser · Mobile · Tablet

Region: Qatar · QAR Currency · Arabic Plate Support

Executive Summary

Why CarTrack Pro — and what it delivers for your business

The Challenge

Car service centres typically manage vehicle tracking, service durations, staff efficiency, and revenue analysis through manual logs, spreadsheets, or disconnected software. This creates critical operational blind spots: delayed service alerts, no vehicle history, zero staff accountability, and missed revenue insights.



Without a connected system, an average shop loses **15–25% of potential revenue** through untracked delays, unbilled add-on services, and staff inefficiency that goes undetected.

The Solution

CarTrack Pro integrates CCTV-based AI plate recognition with a full web management system. Cameras automatically detect plate numbers at entry, match them to customer records, and link data to every service — creating a living database of vehicle history, staff performance, and actionable business intelligence, accessible from any device in real time.



A fully connected system from camera detection → visit creation → service completion → payment → analytics — **with zero manual data re-entry.**

AI AUTOMATION

Zero-touch plate detection

Cameras read and match plates automatically at entry — no staff input needed to identify returning customers.

OPERATIONS

Live service-duration tracking

Start and Done buttons on every service item record exact timestamps — actual duration calculated automatically.

MANAGEMENT

Staff KPIs from real data

Services completed, average duration per staff member, and revenue contribution — all from live system records.

CRM

Complete vehicle history

Every visit, service, staff member, duration, and payment for every vehicle — searchable in seconds.

FINANCE

Revenue & seasonal analytics

Daily, monthly, and seasonal revenue trends. Per-service profitability. Peak-hour identification.

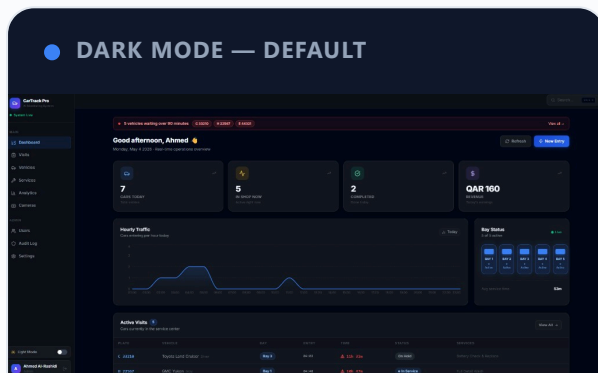
SECURITY

Role-based access + audit trail

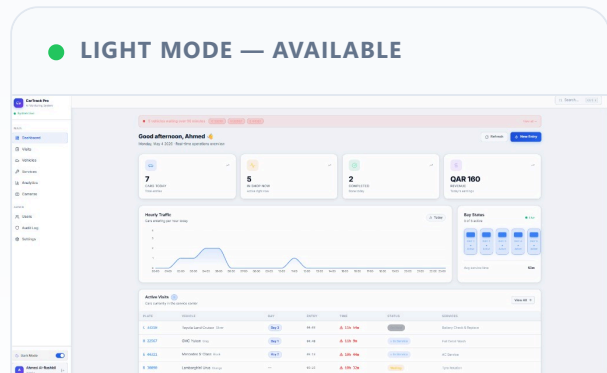
Every create, edit, delete, login, and checkout is permanently recorded. Nothing can be erased.

Dark & Light UI — Built-in Theme System

CarTrack Pro includes a full dark/light mode toggle — switch themes instantly from any page with a single click



Deep navy background with blue accent colours — optimised for extended use in dimly lit service environments and night shifts. Easier on the eyes during long operating hours.

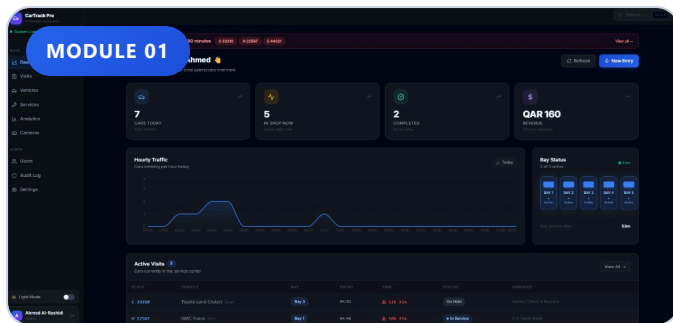


Clean white surface with crisp contrast — ideal for bright reception areas, customer-facing displays, and printed presentations. Theme choice is saved per user device automatically.

i The theme toggle is always accessible from the sidebar footer. The preference is stored in the browser and restored automatically on every login — no configuration required.

Platform Features — Complete Walkthrough

All 11 screenshots are taken from the live production system — every module is fully operational



1

REAL-TIME COMMAND CENTER

Live Operations Dashboard

- Live KPI cards: Cars Today, In Shop Now, Completed, Today's Revenue — refreshed automatically every 30 seconds
- Smart alert banner automatically flags vehicles waiting over 90 minutes with plate numbers and duration
- Interactive 5-bay map shows which bays are occupied live, with average service time displayed per active bay
- Hourly Traffic area chart visualises car entry patterns across the full day to identify peak periods
- Active Visits table with live elapsed-time timers, status colour codes, bay assignments, and service summaries
- Personalised greeting with role-aware content — managers see all data; staff see their own assignments

2

FULL VISIT LIFECYCLE TRACKING

Visit Management

- Status filter tabs — All, Waiting, In Service, On Hold, Completed, Cancelled — with live counts per status
- Live elapsed timers turn amber at 1 hour and red at 2 hours — automatic visual escalation with no manual input
- One-click Checkout, Edit, and Delete directly from each row — zero navigation required for common actions
- Every plate number is a hyperlink to that vehicle's complete history profile
- Real-time search by plate number or customer name — filters as you type across all 50+ records
- Inline delete confirmation modal prevents accidental data loss without leaving the page

The screenshot shows the 'MODULE 02' dashboard. It features a table of visit records with columns: Plate Number, Customer Name, Bay, Status, Elapsed Time, and Actions. The table has a search bar and filter tabs at the top. The status column uses color-coded indicators (green for 'Completed', red for 'Cancelled', and yellow for 'Waiting').

Module 03

Vehicle Info

Cancel

1 Plate Lookup

2 Vehicle Info

3 Services

4 Signature

✓ Returning Vehicle

48 previous visits - Lexus LX600

PLATE NUMBER *

B 98765

ASSIGN BAY

Not assigned

MAKE

Lexus

MODEL

LX600

YEAR

2023

COLOR

Black

VEHICLE TYPE

Suv

CUSTOMER NAME

Fatima Al-Qahtani

PHONE

+974 5512 3402

NOTES

Back

Services

3

INTELLIGENT ENTRY FORM

New Visit — 4-Step Wizard

- Step 1 — Plate Lookup: auto-fills make, model, year, colour, owner name and phone from the database
- Returning-customer banner shows visit count and last visit date — staff see full context immediately
- Step 3 — Services: multi-select services and assign a specific staff member per service item
- Step 4 — Digital Signature: customer signs on screen using touch or mouse before work begins
- Form can pre-fill automatically from an AI camera detection event when a plate is recognised at the gate
- Vehicle type (Sedan, SUV, Pickup, Van, Luxury) recorded at entry for analytics segmentation by type

4

PER-SERVICE PROGRESS MONITOR

Visit Detail & Service Tracking

- Live visit duration counter, total cost (QAR), service count, and payment status in 4 KPI cards at the top
- Each service item has Start and Done buttons — exact timestamps recorded for actual duration calculation
- Actual duration = Done time minus Start time — measured automatically, feeds directly into analytics
- Assigned staff member shown per service with initials avatar — can be reassigned on the fly
- Add Service button allows additional services to be added mid-visit without creating a new entry
- Print Receipt generates a formatted, branded receipt for the customer at checkout

The screenshot displays a mobile application interface for a service appointment. At the top, a dark blue header contains the text "MODULE 04" in white. Below this, the appointment details for "GMC Yukon 2022" are shown, including the date "Monday, 04 May 2026 - 04:48", the location "A Hamad Al-Buainain", and the user "Logged by Ahmed Al-Rashidi". A "Buy 1" button is visible. The main content area features four service cards: "10h 48m" (Duration), "QAR 120" (Total), "1" (Services), and "Unpaid" (Payment). Below these cards, a "Services" section shows the appointment status as "In Progress" for "Hassan Al-Kunari" at "Hamad, 04-48". A "Done" button is present next to the "QAR 120" total. At the bottom, a "Full History" button is located next to the appointment ID "H 22567 - GMC Yukon".

MODULE 04

H 22567 / **in service** HT8158

GMC Yukon 2022

Monday, 04 May 2026 - 04:48 A Hamad Al-Buainain **Buy 1** Logged by Ahmed Al-Rashidi

Print Status **Checkout**

10h 48m DURATION

QAR 120 TOTAL

1 SERVICES

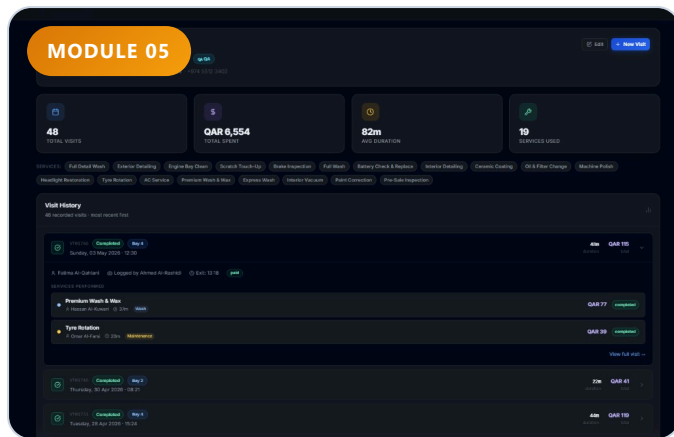
Unpaid PAYMENT

Services [Add Service](#)

Full Detail Wash **In Progress** Wash
Hassan Al-Kunari Hamad, 04-48

QAR 120 **Done**

H 22567 - GMC Yukon [Full History](#)



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COMPLETE CUSTOMER & VEHICLE RECORD

Vehicle Profile & Full History

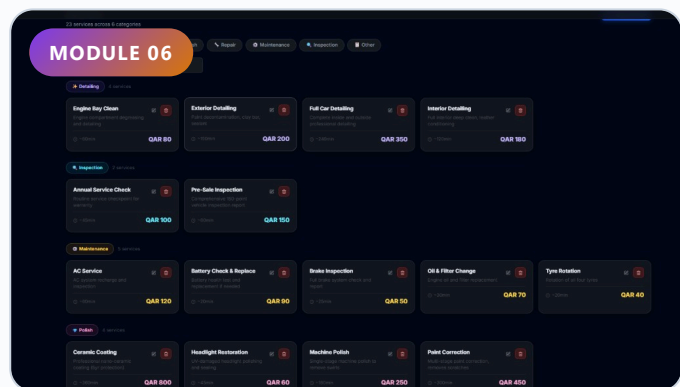
- Header KPIs: Total Visits (48), Total Spent (QAR 6,554), Average Duration (82m), Services Used (19)
- Every service type ever performed on this vehicle shown as colour-coded category chips
- Full chronological visit history — each record shows date, duration, bay, services, staff, and price
- Per-visit service items expanded with staff name, actual duration, and service category tag
- Payment status (Paid / Unpaid / Waived) and bay assignment recorded and displayed per visit
- New Visit button creates a pre-filled entry for this vehicle — one click from the profile

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PRICING & DURATION MANAGEMENT

Services Catalogue

- 23 services across 6 categories: Detailing, Inspection, Maintenance, Polish, Repair, Wash
- Each card: name, full description, estimated duration, and base price in QAR — all editable
- Prices range from QAR 40 (Tyre Rotation) to QAR 800 (Ceramic Coating) covering full service spectrum
- Edit any service — updated price and duration propagate automatically to all new visits
- Category filter tabs for instant navigation — colour-coded per category type
- Estimated duration is used as the analytics benchmark for comparing actual vs. target service times





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BUSINESS INTELLIGENCE & KPI ENGINE

Advanced Analytics — 4 Tabs

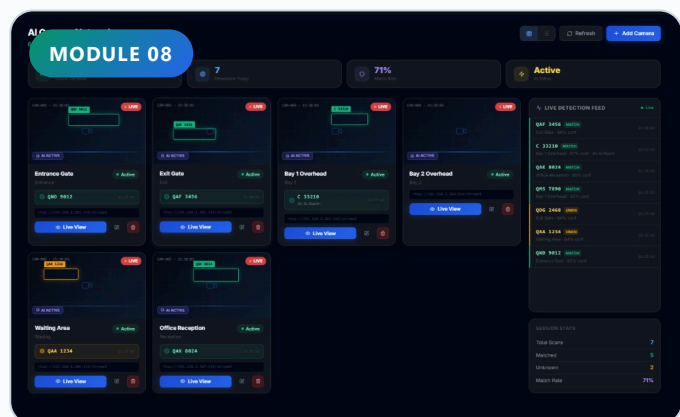
- Overview tab: Total Cars, Avg Duration, Total Revenue, Peak Hour — with trend % vs. previous period
- Daily Traffic & Revenue dual-axis bar chart — see car count and revenue side by side per day
- Service Distribution donut chart — instantly see which service types generate the most volume
- Services tab: actual vs. estimated duration per service type — identify where time overruns occur
- Staff KPIs tab: services completed, average duration per job, and revenue attributed per staff member
- Seasonal tab: full-year monthly revenue and avg duration — plan staffing and pricing by season

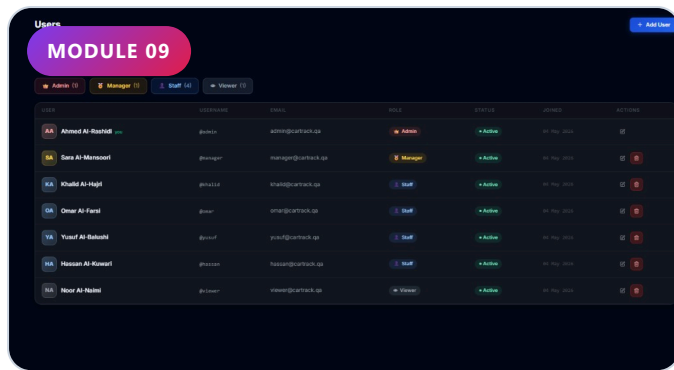
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LIVE VEHICLE DETECTION & PLATE RECOGNITION

AI Camera Network

- 6 live camera feeds — Entrance Gate, Exit Gate, Bay 1–5, Waiting Area, Reception
- Each camera shows a live bounding-box overlay on the detected vehicle with plate label
- Green labels = MATCHED known customers; Amber labels = UNKNOWN unregistered plates
- Confidence score (e.g. 94%) shown per detection — based on YOLOv8 + EasyOCR confidence
- Live Detection Feed sidebar: real-time stream of all detection events across all cameras
- Session Stats panel: Total Scans, Matched, Unknown, Match Rate — updated live per session





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ROLE-BASED ACCESS MANAGEMENT

Users & Access Control

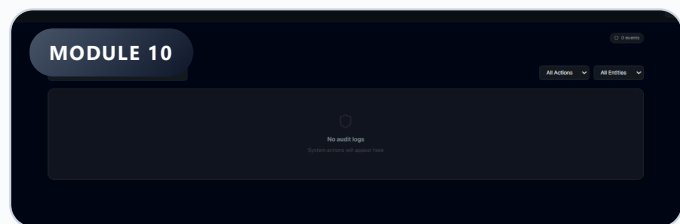
- 4 roles: Admin (full access), Manager (operations + reports), Staff (visits + services), Viewer (read-only)
- Admin and Manager roles unlock the full Admin section — Users, Audit Log, Settings
- Each user: avatar with initials, full name, username, email, role badge, status, and join date
- Filter by role using the pill tabs at the top — see all Staff, all Managers, etc. instantly
- Edit user details or reset passwords from the action menu — no backend access needed
- Status badge (Active / Inactive) — deactivate accounts without deleting their history

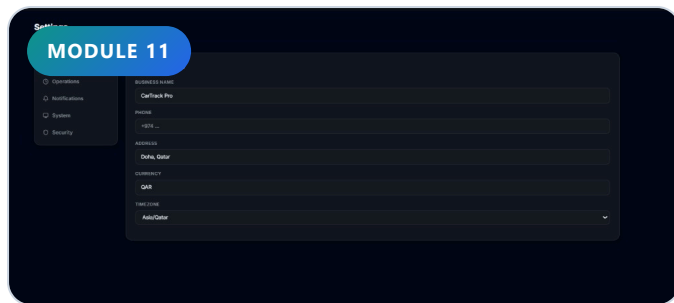
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FULL COMPLIANCE & ACTIVITY TRAIL

Audit Log

- Every create, update, delete, login, logout, and checkout is permanently recorded with timestamp
- Timeline view: user avatar, colour-coded action badge (Create / Update / Delete), entity, and time
- Filter by action type and entity type independently — narrow down to specific events instantly
- Pagination at 40 events per page — loads fast even with thousands of historical records
- Immutable — no user at any role level can delete or modify audit entries
- Essential for insurance audits, dispute resolution, and staff accountability reviews





11

FULL CONFIGURATION CONTROL

System Settings

- General: Business name, phone number, address, currency (QAR default), and timezone (Asia/Qatar)
- Operations: Service bay count, peak hour window definition, and late-car alert threshold (minutes)
- Notifications: Configure alert triggers for late vehicles and daily summary report delivery
- System: Backup schedule, data retention period, and external API integration management
- Security: Password complexity rules, session timeout, and two-factor authentication toggle
- 5-section sidebar navigation — General, Operations, Notifications, System, Security

AI Camera Technology Stack

How the system sees, reads, matches, and responds to every vehicle in real time

STEP 1 — VEHICLE DETECTION

YOLOv8 Object Detection

YOLOv8 processes each camera stream at 15–30 fps on the GPU. It draws a bounding box around every vehicle detected in under 50ms per frame. The model is pre-trained on 80+ object categories and fine-tuned for vehicle detection in the Qatari environment.

STEP 2 — PLATE RECOGNITION

EasyOCR Plate Reading

The cropped vehicle region is passed to EasyOCR, which reads the plate text in milliseconds. Qatar plate formats — including Arabic and Latin characters — are supported. Accuracy reaches 85–97% in good lighting with a 4MP+ camera within 6 metres of the plate.

STEP 3 — DATABASE MATCHING

Real-Time Vehicle Lookup

The detected plate string is compared against all vehicles in the database via a fast API query. A match triggers a green overlay with owner name and visit count. No match triggers an amber UNKNOWN flag, prompting staff to register the new vehicle.

Full Detection-to-Analytics Data Flow

- 1 IP Camera streams live RTSP video to the AI processing server over the local network
- 2 YOLOv8 runs on the GPU — detects vehicles and creates bounding boxes per frame at 15–30 fps
- 3 EasyOCR reads the plate number text from the cropped bounding box region
- 4 FastAPI backend queries the PostgreSQL vehicle database for a plate match
- 5 Detection event (plate, match status, confidence, camera ID) sent to all browsers via WebSocket
- 6 Live Detection Feed sidebar in the Camera page updates in real time — staff see the event immediately
- 7 Staff clicks "Create Visit" — form pre-fills with vehicle data from the matched detection event
- 8 Services selected, staff assigned, signature captured — visit record created and linked to vehicle
- 9 On Done — actual service duration measured, revenue recorded, analytics dashboard updated live

Recommended Camera Hardware

Specifications ensure 90%+ plate recognition in Qatar's climate — sun glare, dust, and 24/7 operation

Resolution

4–8 MP

Minimum 4MP (2560×1440). Recommended 8MP/4K for gate cameras reading plates at over 5m.

Frame Rate

25–30 fps

Required for moving vehicles at the gate. 15 fps minimum for static bay monitoring.

WDR

120 dB+

Wide Dynamic Range handles Qatar sun glare, headlights, and high contrast at entry/exit.

Night Vision

Protocol

IP Rating

IR 50m+ Smart IR up to 50m with auto intensity adjustment for 24/7 operation without colour washout.	RTSP/ONVIF Mandatory for AI server integration. ISAPI (Hikvision) or SDK access preferred for advanced control.	IP67 Full dust-tight and water-resistant housing for outdoor Qatar conditions including summer sandstorms.
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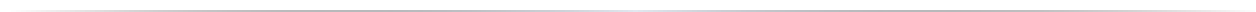
Recommended Camera Models (Qatar Market)

DS-2CD2685G1-IZS	Hikvision	ANPR Bullet	8MP Varifocal	Entrance / Exit gate — primary ANPR
DS-2CD2T47G2-L	Hikvision	ColorVu Bullet	4MP Full-colour	Entrance / parking — colour night vision
DS-2CD2147G2-LU	Hikvision	ColorVu Dome	4MP Full-colour	Service bays — overhead vandal-proof dome
IPC-HFW5842H-ZAS	Dahua	ANPR Bullet	8MP Varifocal	Entrance / Exit gate — Dahua alternative
IPC-HDW3849H-AS-PV	Dahua	Full-Colour Dome	4MP	Indoor bays, reception, waiting area
DS-9632NI-I8 NVR	Hikvision	32-Ch NVR	4K H.265+	Central recording unit — RAID storage backup

Camera Placement Guide

LOCATION	CAMERA TYPE	PURPOSE	HEIGHT	ANGLE
Main Entrance Gate	8MP ANPR Bullet	Plate capture on vehicle entry — primary detection point	3–4m	15–25° down
Exit Gate	8MP ANPR Bullet	Plate capture on exit — confirms checkout and exit time	3–4m	15–25° down
Service Bay 1–5	4MP Dome Overhead	Vehicle and technician activity monitoring per bay	4–5m	90° vertical
Waiting Area	4MP Dome Wide	Customer and vehicle queue overview	3m	45° sweep
Reception / Cashier	2MP Dome	Transaction security and audit recording	2.5m	Wide angle
Workshop Overview	4MP Bullet	Full workshop bird's-eye view	5–6m	Wide angle
Outdoor Parking	4MP Varifocal Bullet	Vehicle queue and incoming detection	4m	20° down


Installation Note: ANPR cameras must be mounted no more than 4–6 metres from the plate read point. The lens axis should remain within 30° of horizontal, and the plate should occupy at least 25% of the frame width for reliable OCR. Avoid backlit or west-facing mounts during sunset hours. Heating shrouds are recommended for outdoor positions in Doha's summer months (June–September).



Full System Requirements

Hardware, software, and network specifications for a production-ready deployment

AI Processing Server — On-Premise

Component	Minimum	Recommended
CPU	Intel Core i7-12700 / AMD Ryzen 7 5800X	REC Intel Core i9-13900K / AMD Ryzen 9 7950X
RAM	32 GB DDR4	REC 64 GB DDR5
GPU (AI Inference)	NVIDIA RTX 3060 12GB VRAM	REC NVIDIA RTX 4080 / RTX 4090
Primary Storage	1 TB NVMe SSD	REC 2 TB NVMe SSD (OS + models + database)
Video Archive	8 TB HDD RAID-1 (30-day retention)	REC 16 TB RAID-5 NAS (90-day retention)
Network Card	1 Gbps Ethernet	REC 2.5 Gbps Ethernet
UPS	1500VA APC or equivalent	REC 3000VA with AVR and auto-shutdown
Operating System	Windows Server 2022	REC Ubuntu 22.04 LTS (preferred for AI stack)

Network Infrastructure

Component	Specification	Purpose
PoE Switch	24-port Managed PoE+ (802.3at), Gigabit	Power and connect all IP cameras via single cable run
Core Switch	Managed L2/L3 Gigabit with VLAN support	Isolate camera VLAN from office and client networks
NVR (backup)	32-channel 4K NVR with RAID storage	Local video recording independent of AI server
Internet	100 Mbps symmetric fibre (minimum)	Remote management, cloud backup, and software updates
Firewall / VPN	FortiGate 60F / pfSense with IPS	Secure remote access and network segmentation
Wi-Fi	Wi-Fi 6 AP — Ubiquiti / TP-Link Omada	Staff tablets, mobile access, customer-facing displays

Software Stack

LAYER	TECHNOLOGY	ROLE
AI Engine	Python 3.11, YOLOv8, EasyOCR, OpenCV	Vehicle detection and plate reading from RTSP streams
Backend API	FastAPI (Python), SQLAlchemy ORM, Alembic	REST API serving all frontend requests and business logic
Database	PostgreSQL 15 (production deployment)	Persistent storage for visits, vehicles, users, analytics
Real-time Push	WebSocket — FastAPI native	Camera detection events and live updates pushed to frontend
Frontend	React 18, TypeScript, Vite, TanStack Query	Full web dashboard — all 11 modules, dark + light themes
Charts	Recharts	All analytics visualisations — bar, line, pie, donut, area
Authentication	JWT tokens + bcrypt, RBAC middleware	Secure login, persistent sessions, role-based access
Containerisation	Docker + Docker Compose	One-command deployment, environment consistency
Reverse Proxy	Nginx with HTTPS/TLS	SSL termination, static asset caching, load balancing

Client Device Requirements

DEVICE	MINIMUM SPEC	PRIMARY USE CASE
Manager Workstation	Any modern PC, 8 GB RAM, Chrome / Firefox / Edge	Dashboard, analytics, user management, audit review
Reception Tablet	10" iPad or Android tablet, Wi-Fi, Chrome	New visit wizard, customer digital signature capture
Staff Mobile	iOS 14+ / Android 11+, any modern browser	Start/Done service buttons, visit status updates
Bay Display	24"+ HDMI monitor, connected to AI server or PC	Current vehicle, service status, duration for technicians

Implementation Phases

A structured 6-week go-live plan followed by ongoing optimisation and support

Weeks 1–2

Core Infrastructure & Installation

- AI server hardware procurement, rack installation, and Ubuntu OS configuration
- Network switch, structured cabling, PoE layout, and camera physical installation
- PostgreSQL database setup, Alembic schema migration, and initial admin data load
- FastAPI backend and Nginx reverse proxy deployment with HTTPS/TLS certificate
- React frontend build and static file deployment — accessible on local network

Weeks 3–4

AI Integration & Accuracy Calibration

- YOLOv8 model calibration per camera angle, lens zoom, and Qatar lighting conditions
- EasyOCR fine-tuning for Qatar plate format — Arabic numerals and Latin characters
- RTSP stream integration — all cameras connected and verified in the system
- WebSocket detection event testing — live feed verified in the Camera Network module
- Plate recognition accuracy validation — target 90%+ across all gate cameras

Weeks 5–6

Staff Training & Operations Go-Live

- Full staff training on visit wizard, service tracking, checkout, and signature capture
- Historical vehicle and customer data import from existing records
- Service catalogue setup — prices, descriptions, and estimated durations configured
- Analytics dashboard review with management — KPI baselines established
- Receipt printing, mobile access, and tablet signature-capture testing and sign-off

Ongoing

Support, Optimisation & Growth

- Monthly AI model accuracy reports — re-training as plate formats or conditions change
- Feature additions based on operational feedback and new business requirements
- Database backup verification and disaster recovery drill (quarterly)
- Performance monitoring, scaling, and security patch management
- Analytics review sessions — revenue optimisation insights delivered to management

System Value at a Glance

Live figures from an existing client implementation — recorded over the first operational weeks of the system

49 CARS (7 DAYS) +12% vs prev	79m AVG DURATION -5% improved	QAR 7,128 REVENUE (7 DAYS) +18% vs prev	10:00 PEAK HOUR Identified	71% AI MATCH RATE Live cameras
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